

软件学院 2023 级软件工程专业 (2023-2)

《计算机网络》 期末试题试卷(B)

(考试形式：闭卷 考试时间：2 小时)



《中山大学授予学士学位工作细则》 第六条

考试作弊不授予学士学位

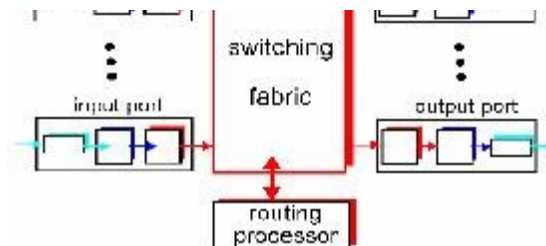
方向: _____ 姓名: _____ 学号: _____ 成绩: _____

注意：答案一定要写在答卷中，写在本试题卷中不给分。本试卷要和答卷一起交回。

答卷指南：

- 1) 考题共 8 题，另有一道加分题。
- 2) 可选中文或者英文来答卷。

- 1) (10points) Compute the CRC for a given message (M) and a generator polynomial (P). M is 0111101 and P is 1011.
- 2) (10points) CSMA/CD
 - a) Explain CSMA/CD protocol, and how it backs off when conflicts happen.
 - b) Compute the minimum possible frame size for a CSMA/CD protocol given the following parameters. Maximum medium span is 5000 meters (signal propagation is 5 nanoseconds per meter) and the data rate is 100-Mbps.
- 3) (10points) Where (input ports and/or output ports) can queueing occur in a router? Briefly explain the conditions that lead to such queueing.



- 4) (15points) Explain the distance vector routing algorithm and give answers to the following questions.

A router has the following (CIDR) entries in its routing table:

Address/mask

Next hop

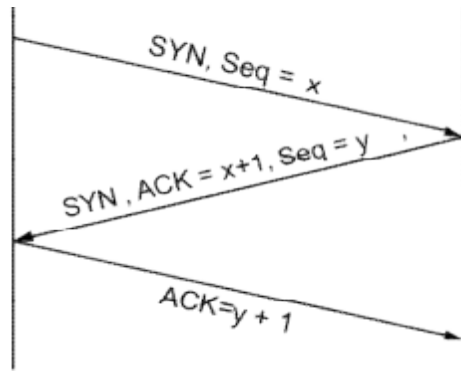
Address/mask	Next hop
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

- (a) 135.46.63. 10
- (b) 135.46.57. 14
- (c) 135.46.52.2
- (d) 192.53.40.7
- (e) 192.53.56.7

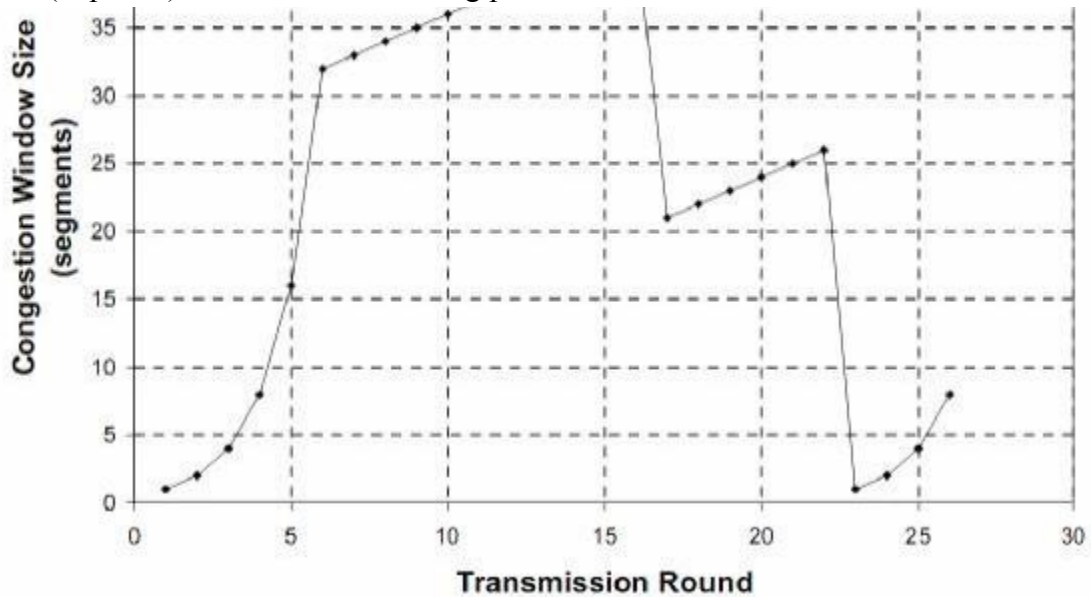
- 5) (15points) Consider a 4-layer protocol implementation with application, TCP, IP, and Ethernet layers in that order (top to bottom). Each layer requires a header except the Ethernet layer, which requires a header and trailer. The application header is 16 bytes in length, TCP header 20 bytes, IP header 20 bytes, and let the Ethernet header be 14 bytes, and the trailer 4 bytes (ignore the preamble and gap). Answer the following questions:
- a) Sketch a packet for this system carefully showing and labeling all fields.
 - b) Assume a maximum data field for an Ethernet frame of 1600 bytes. What is the overhead (in %) for a 4096-byte application message? Hint: the message must be segmented into multiple frames and be careful of how you consider the data field in the Ethernet frame.
- 6) (10points) Consider in the Transport Layer, clients and servers use SYN, ACK, FIN etc to initiate and terminate connections.

When initiating a connection, the following sequence of SYN/ACK takes place.



Please describe how a web browser terminates its connection with a web server by following the example above.

7) (15points) Consider the following plot of TCP window size as a function of time:



Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions.

- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating(AIMD)
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?

- 8) (15points) CSMA/CA
- What is CSMA/CA?
 - Why WIFI uses CSMA/CA?
 - How CSMA/CA works?

Bonus Question: (10points) Suppose a system picks a symmetric key authentication protocol using an authentication server. Below $\{m\}_k$ denotes the encryption of message m by key k . We assume A is a trusted authentication server which shares a secret key k_{XA} with each principal X in the system.

(1) $P \rightarrow Q$: "I am P."
 (2) $Q \rightarrow Q$: generate nonce n
 (3) $Q \rightarrow P$: n
 (4) $P \rightarrow Q$: $\{n\}_{k_{PA}}$
 (5) $Q \rightarrow A$: $\{P, \{n\}_{k_{PA}}\}_{k_{QA}}$
 (6) $A \rightarrow Q$: $\{n\}_{k_{QA}}$

Someone has identified a serious flaw in the protocol. Can you identify the step(s) which may cause security problems? Can you propose a fix?